Ambient IoT in 3GPP Release 19: A Survey

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***Abstract*—The ambient Internet of Things (IoT) that utilizes harvest energy is gaining recognition as an innovative technology in the wireless communication industry. The devices associated with this technology, known as Ambient IoT devices, are characterized by being battery-less, low-cost, maintenance-free, smaller size, and low hardware complexity. Recently, the Third Generation Partnership Project (3GPP) has initiated a study on integrating Ambient IoT with the cellular network, to provide widespread and flexible connectivity, meeting the massive deployments of these class ultra-low power Ambient IoT devices. This paper outlines a timely survey of the Ambient IoT within 3GPP Radio Access Network (RAN) Release 19. It presents the technical foundations of Ambient IoT, encompassing representative use case, connectivity typology, service requirement, and device type. Furthermore, it delves into an analysis of the essential protocol functions and gives the signaling procedures for the Ambient IoT. In addition, the paper explores several potential considerations for Ambient IoT that might be explored in the future.**

***Index Terms*—Ambient IoT, harvest energy, battery-less, low-cost, maintenance-free, 3GPP.**

I. Introduction

The rapid expansion of IoT devices has seen substantial growth in recent years, with the total number projected to reach an astounding 34.7 billion by 2028, as indicated from [1]. Meanwhile, the IoT applications encompass a wide array of vertical markets [2] [3], such as smart city, smart transportation, smart energy, smart factory, etc. It is important to note that the increasing prevalence of IoT devices and their widespread connectivity bring some new challenges for future wireless networks, particularly in terms of energy consumption and safety challenges. Addressing these issues, [4] and [5] point out that energy harvesting technology plays a pivotal role in addressing the high maintenance cost in the further digital IoT industry. More specially, replacing manual charging or powered by batteries for IoT devices, the energy harvesting enabled IoT devices can obtain energy from the environment (i.e., radio frequency energy, solar energy, thermal energy, and electromagnetic energy), which greatly increases the lifespan of IoT devices and reduces power costs. In addition, [6] proposes a collaborative mutation-based moving target defense (CM-MTD) algorithm, to avoid and mitigate the security threats posed by a large number physical IoT devices in the digital twins mobile network (DTMN).

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In the past, the 3rd Generation Partnership Project (3GPP) has conducted extensive research and developed specifications for the cellular solutions for the various classes of IoT devices. In long-term evolution (LTE), the narrow band IoT (NB-IoT) and LTE for machine-type communications (eMTC) were first introduced in Release 13, which are used to support cellular IoT for a variety of deployment scenarios, including improved indoor coverage, support of a massive number of low-throughput devices, reduced complexity, and improved power efficiency [7]. Reduced capability (RedCap) technology based on fifth-generation mobile networks (5G) was officially approved as a study item in Release 17, and the NR RedCap devices are proposed to expect low-cost and low power consumption compared with LTE IoT devices [8]. In addition, the significant advantage of RedCap is that supporting the coexistence with enhanced mobile broadband (eMBB) and low-latency communication (URLLC) on a new radio (NR) carrier.

However, the traditional cellular IoT technologies investigated by 3GPP still employ battery-powered IoT devices. Considering the increasing interest in energy-efficient networks for the IoT market, the next generation of IoT device is anticipated to evolve towards being battery-less or with limited energy storage capability [9]. In February 2022, 3GPP launched a study of Ambient IoT in the Service and System Aspects working group 1 (SA1), which describes use cases, traffic scenarios, and device constraints of ambient IoT. Additionally, it also identifies some new potential service requirements and key performance indicators (KPIs). Details of this study are captured in Technical Report (TR) 22.840 [10]. Subsequently, the Technical Specification (TS) 22.369 has further clarified and specified the service and performance requirements for Ambient IoT [11]. Additionally, at the SA2, a new study on the architecture support of Ambient IoT devices is also approved, which is based on the specified service requirements and documented in TR 23700-13 [12]. Within the 3GPP Radio Access Network (RAN) community, a study item (SI) [13] about Ambient IoT has been approved in Release 18, with the primary aim is to investigate the feasibility of meeting the design targets for relevant use cases of a new 3GPP IoT technology. The RAN plenary study has approved the new study item (SI) RP-240826 [14] titled "Study on solutions for Ambient IoT (Internet of Things) in NR" in Release 19, which

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TABLE I
SERVICE REQUIREMENTS FOR REPRESENTATIVE USE CASE

Use case	Data rate	Latency	Availability	Reliability	Message size	Communication range	Positioning accuracy
Inventory (indoor)	$<$ 2 Kibts/s	Seconds level	99	NA	96/256 bits	30-50 m	3 m
Inventory (outdoor)	$<$ 2 Kibts/s	Seconds level	99	NA	96/256 bits	200-400 m	Cell level
Command (indoor)	$<$ 2 Kibts/s	Seconds level	99	NA	100Bytes	50 m	3-5 m
Command (outdoor)	$<$ 2/NA Kibts/s	Seconds level	99	NA	128/200bits	200-500 m	NA

overall objective is to study a harmonized air interface design with minimized differences for the different Ambient IoT deceives crossing multiple deployments scenarios. The corresponding document TR 38.769 [15] offers potential solutions from different WG groups according to the defined device types, traffic types, use cases, and connectivity typologies.

In this paper, we present a comprehensive survey of the Ambient IoT in the latest 3GPP Release 19. The paper is organized as follows. In section 2, we give a brief technical background of the Ambient IoT device. The analysis of protocol aspects is explained in section 3. Section 4 describes the general signaling procedures for the approved Ambient IoT cases. Finally, in section 5, we share some potential considerations related to Ambient IoT for future enhancements.

II. TECHNICAL BACKGROUND

The technical research work of Ambient IoT is ongoing in 3GPP, we provide a concise and compact summary of the technical background cited in [10-15], involving representative use case, connectivity topology, service requirement, and device type.

- **Representative use case**

Representative use cases of Ambient IoT can be categorized based on deployment environment or functionality, we list the mainly typical use cases on the basis of their services.

- Inventory: The Ambient IoT supports inventory services with the purpose is to discovering the goods that are present in a specific area, such as automated warehousing, medical instruments inventory management and positioning, intralogistics in automobile manufacturing, and etc. From standardization prospective, it refers to as a procedure used to discover and acquire the identifier of single or multiple Ambient IoT devices.

- Command: The Ambient IoT supports various control management services. For instance, in the scenario of smart grazing dairy farming enabled by Ambient IoT, the Ambient IoT device can be controlled to update its status or data information. It can be described that the network sends an operation request (i.e., write, read, enable, disable) to the single or multiple Ambient IoT devices within the standardization scope.

- **Connectivity topology**

There are four connectivity topologies specified and identified in 3GPP, notify that the carrier wave (CW) nodes for the Ambient IoT device does not described in the topologies. In addition, Release 19 only focuses on the research and investigation of both Topology 1 and Topology 2.

- Topology 1: BS $\longleftrightarrow$ Ambient IoT Device;

- Topology 2: BS $\longleftrightarrow$ intermediate node $\longleftrightarrow$ Ambient IoT Device;

- Topology 3: BS $\longleftrightarrow$ intermediate node $\longleftrightarrow$ Ambient IoT device$\longleftrightarrow$ BS;

- Topology 4: UE $\longleftrightarrow$ Ambient IoT device.

In Topology 1, Ambient IoT device support the direct and bidirectional communication with a base station (BS). During Topology 2 and Topology 3, the Ambient IoT device requires an intermediate node to assist its communication with BS, which is beneficial for extending the coverage area in comparison to Topology 1. It needs to be noted that the primary distinction between Topology 2 and Topology 3 is the communication direction, which is bidirectional in Topology 2 and unidirectional in Topology 3. In addition, the intermediate node can be a relay, IAB node, UE, repeater, etc. which is capable of Ambient IoT. In regards to Topology 4, an Ambient IoT device engages in bidirectional communication with a UE. Moreover, in 3GPP, the node that directly communicates with Ambient IoT devices is referred to as a reader including the UE reader and the BS reader.

- **Service requirement**

Regarding extremely limited power constraint, ultra-low hardware complexity of the Ambient IoT, the service requirements of Ambient IoT are summarized in Table 1. It depicts the specific requirements in terms of data rate, latency, availability, reliability, message size, and communication range for the Inventory and Command use cases in different deployment scenarios (indoor and outdoor). From Table 1, it can be observed that the service requirements are a little bit different among various use cases. In fact, the Ambient IoT is designed to support these requirements which are not extreme in terms of data rate, latency, and reliability. Instead, Ambient IoT is optimized for supporting the low-end network, so a significant focus is given to the lifespan of Ambient IoT devices, which typically is anticipated to be around 10 years.

- **Device type**

Ambient IoT devices possess the limited energy storage capability. The standardization has approved the following classification of Ambient IoT devices relying on their respective energy storage capacities and generating RF signals capabilities.

- Device A: No energy storage, no independent signal generation/amplification, i.e. backscattering transmission.

- Device B: Has energy storage, no independent signal generation, i.e. backscattering transmission. Use of stored energy can include amplification for reflected signals

- Device C: Has energy storage, has independent signal generation, i.e., active RF components for transmission.

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III. Functions of Protocol Stack

The existing New Radio (NR) protocol stacks, as provided in [16], consist of a control plane and user plane over the air interface. More precisely, the control plane is comprised of a Radio Resource Control (RRC) layer, Packet Data Convergence Protocol (PDCP) layer, Medium Access Control (MAC) layer, and Physical layer. On the other hand, the user plane consists of the Service Data Adaptation Protocol (SDAP), PDCP layer, MAC layer, and Physical layer. However, a consensus has been reached to decrease the design complexity of the protocol stack for Ambient IoT comparing to the existing NR protocol stacks. This decision is based on the substantial reduction in projected peak power consumption and hardware complexity of Ambient IoT, in comparison to current 3GPP IoT technologies. Given these characteristics, it is evident that an Ambient IoT device may only need to support a limited set of functionalities. In order to simplify this process, the 3GPP RAN is actively involved in conducting research to identify a more lightweight and efficient protocol stack for Ambient IoT. Moreover, it involves a thorough investigation into the core protocol functions that are critical for smooth operation of Ambient IoT devices, which is to ensure that the new protocol stack not only meets the unique requirements of Ambient IoT devices, but also maintains the necessary functionalities for effective communication.

check what each layer isresponsible for

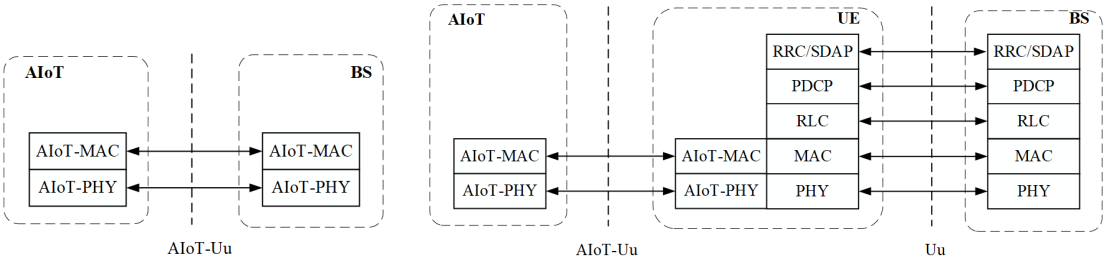
simplify the protocol stack without losing necessary fynctionality

RRC layer is the primary control layer that configures and manages other control layers at the AS level. The traditional NR RRC layer serves a multitude of functions including broadcast of system information, paging, RRC connection management, mobility functions, measurement configuration and reporting, Quality of Service (QoS) management, non access stratum (NAS) message transfer. While in the Ambient IoT system, the aforementioned functions seem to be unnecessary and superfluous. It is well known that the minimum system information (MSI) including MasterInformationBlock (MIB) and SystemInformationBlock1 (SIB1) is used to the initial network access, which is broadcasted to UEs in a periodic manner. However, for the current Ambient IoT traffic types, both Device-terminated (DT) and Device-originated - device-terminated triggered (DO-DTT) are initiated by the network and sent to UE. Observably, there is no essential information has to be acquired for the Ambient IoT device.The ultra-low complexity Ambient IoT device may also feature a sample frequency offset (SFO). Therefore, a synchronised network may not to be needed for Ambient IoT device, since it is hard to configure common parameters for various kinds of Ambient IoT devices. Furthermore, the dedicated physical broadcast channel has already been already excluded during RAN 1 discussion. Therefore, the broadcast of system information is unnecessary in Ambient IoT system. According to the RAN1 study, it was agreed that Ambient IoT devices do not have RRC states. As a result, mobility management functions like as cell selection, cell re-selection, and handover function are not supported. Meanwhile, it is natural that the measurement configuration and reporting framework is also not needed since it is used to mobility management. There is no significant difference for different Messages of Ambient IoT, so no justification for implementing QoS management functionalities. Additionally, paging involves the network signaling to a UE that it needs to take some action, such as waking up from a low-power(i.e., RRC IDLE, RRC INACTIVE) state to receive signaling data. Based on the above discussion, it is clear that Ambient IoT does not need this functionality, but the transmission mechanism can be retained and reused for other Ambient IoT -related information, such as scheduling information. Similarly, NAS Transfer means that the NAS message is contained in RRC messages and be transmitted between the UE and the network. For the Ambient IoT, if there is corresponding higher-layer information may be existent, which also needed be carried in the lower layer for the end-to-end transmission. For the Ambient IoT, if there is some corresponding higher-layer information, it is necessary to transmit this information at a lower layer in order to ensure end-to-end transmission. Hence, to reduce the burden of Ambient IoT device, such mechanism (i.e., paging-like, NAS-like transfer) can be directly supported in MAC layer. Hence, the RRC layer seems to be unnecessary for Ambient IoT.

The SDAP plays a critical role in the 5G NR by bridging the gap between the application layer (QoS flow) and the lower layer (Data Radio Bearers, or DRBs). Its primary function is the mapping between QoS flows and DRBs to ensure that the QoS requirements of different data traffics. However, this function of the SDAP layer may be redundant, since there is no traditional Protocol Data Unit (PDU) session and QoS flow concept for Ambient IoT. NR PDCP layer performs several essential functions such as header compression, AS security which includes cyphering and integrity protection, among others. However, in the Ambient IoT, the necessity of these functions may be reconsidered. Ambient IoT typically involves small data transmissions with a maximum message size of approximately 1000 bits. Header compression is mostly beneficial for large packets, like IP packets. Thus, this feature may not yield substantial advantages for Ambient IoT traffic. On the other hand, the 3GPP SA3 is in charge of the discussion on the necessity of AS security. Nevertheless, in order to reduce Ambient device complexity, 3GPP RAN has assumed that the security will not be considered at the AS layer. The primary role of the RLC layer is to perform Automatic Repeat Request (ARQ), as well as segmentation and reassembly. By conducting the Ambient IoT study without employing Automatic Repeat Request (ARQ), the necessity of include duplicate detection for Acknowledged Mode (AM) in the RLC layer is excluded. Regarding segmentation and reassembly, it is worth noting that data can be efficiently conveyed at once because the maximum data size is not too large. Additionally, segmentation can be performed at the physical layer if needed,. As above analysis, there is no need to introduce the SDAP layer, PDCP layer and RLC layer for Ambient IoT.

NR MAC layer supports a number of important functions, including random access, scheduling information reporting, error correction through hybird automatic repeat request

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(a) Ambient IoT protocol stacks for topology 1

(b) Ambient IoT protocol stacks for topology 2

Fig. 1. Simplified protocol stacks for Ambient IoT

(HARQ), multiplexing and demultiplexing, discontinuous reception(DRX) and so on. In the Ambient IoT, MAC layer has the responsibility for handling the random access procedure and to provide services between higher layers (e.g. NAS) and lower layer (e.g. physical layer), and the slotted-ALOHA based access was regarded at a baseline to study for the data transmission at current stage. Both Scheduling Request (SR) and Buffer Status Report (BSR) mechanisms serve the purpose of notifying the network about the presence of additional data ready for transmission from the UE side. More specifically, the SR is used to indicate when the UE has the data but there is no uplink grant. However, in the Ambient IoT, the Ambient IoT device can transmit the data or signaling only when receiving an indication for data transmission from network. Thus, the SR is not needed for Ambient IoT. While the BSR may be still beneficial to leading to efficient utilization of network resource in the Ambient IoT system. For example, Ambient IoT device may optionally send an implicit or explicit indication to the network to indicate the total size of buffered data. However, in order to decrease the devices complexity and avoid additional power consumption of Ambient IoT, the study rules out the use of HARQ, which can be replaced by blind re-transmission in Ambient IoT. Moreover, it is no need to consider the multiplexing and demultiplexing since there may be only one logical channel, or the transmission of signaling and the transmission of data are still usually separate if have two logical channels. While the DRX is widely used to conserve the power consumption by monitoring the PDCCH at specific time duration. However, However, Ambient IoT (A-IOT) devices are typically engaged in transmitting small data payload infrequently and does not need multiple times transmission, so introducing DRX for Ambient IoT may not significantly benefit compared to increased complexity. Consequently, it is indispensable to tailor the MAC layer to align with the functionalities required by Ambient IoT systems for efficient data transmission. This specifically involves integrating features such as random access mechanisms and a simplified BSR. AIoT-pHY are included in the Ambient IoT protocol stacks

Figure 1 shows the possible protocol stacks including AIoT-MAC layer and AIoT-PHY layer for Ambient IoT in different typologies. Figure 1 (a) describes the Ambient IoT protocol stack for topology 1, and Figure 1 (b) depicts the Ambient IoT protocol stack for topology 2.

IV. General Signaling Procedures

3GPP has agreed that only focuses on Inventory and Command use cases in Release 19. Accordingly, RAN also began to initiate researching into stage 2 signaling procedures and information to support the proposed use cases. Due to the characteristics of Ambient IoT devices (i.e., minimal hardware complexity, and limited power consumption), the signaling procedures need to be simplified during the study.

A. Overall signaling procedures

In the beginning, RAN 2 addressed three cases for the overall signaling procedures, encompassing Inventory-only, Inventory-Command, and Command-only. For Command-only case, if the UE reader or BS reader wants to send a command-related signaling to a single device, it could skip the access and perform the command phase directly, which means that the command information (e.g., device ID) is included in the initial triggering message. It can be easily observed that this way is able to reduce the signaling overhead and improve the signaling efficiency. However, this may provide a security risk because it is unclear if the CN or reader can maintain the security context. Thus, RAN2 cannot address the security issue directly associated with the command-only scenario until a conclusion is determined during the discussions in SA2 and SA3. On the contrary, it is agreed that the RAN domain primarily addresses the overall signaling procedures exclusive to cases involving either Inventory-only or Inventory-Command cases. These cases require cooperation with the SA2, emphasizing the importance of taking a comprehensive approach to system architecture and coordinating procedures. The primary aim from the RAN perspective is to precisely delineate and identify the interactions between the reader and the Ambient IoT device.

Figure 2 illustrates the overall signaling procedure of Inventory-only. The inventory request message is generated CN functions such as Access and Mobility Management Function (AMF) or a new CN function for Ambient IoT, possibly named Ambient IoT function (AIoTF). As shown in Figure 2, the detailed procedure flows are described as follows.

Step 1. The CN function sends a inventory request message to the reader to facilitate the reader to perform inventory requests for the Ambient IoT device. This high layer message includes some information about requested Ambient IoT device, such as identifier (ID) information and criteria

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information. Moreover, the inventory request message can be sent in a periodic manner.

Step 2. The reader sends a paging-like message to the Ambient IoT device, and the device information (e.g., temporary ID or permanent ID) is included in the paging-like message. In addition, the message also carries the radio resource assignment information.

Step 3. The Ambient IoT device initiates a random access procedure to transmit its data information. During the random access procedure, a new random ID is needed for the contention resolution purpose.

Step 4. The Ambient IoT device sends the inventory response including its device ID to the reader after random access is successfully performed.

Step 5. Based on the response message received from the Ambient IoT device, the reader forwards the inventory response to the CN. It is noted that some information in the inventory response message may be visible to the reader.

Figure 3 depicts the comprehensive signaling process of Inventory-Command, where the Command operation consistently succeeds the Inventory procedure. The command request message, similar to Figure 2, is created by CN functions such as AMF or AIoTF. The detailed procedure flows are described below.

Step 1. The reader receives a command request message from CN functions, which is used to indicate the reader to perform a command request (i.e., read, write, enable, disable)for the Ambient IoT device. The Ambient IoT device-related information is carried in the command request message. Its signaling transmission scheme is the same to the command request message, which is sent periodically.

Step 2. The inventory operation is performed between the Ambient IoT device and the reader, which includes the paging-like procedure and random access procedure, which is intend to inventory the requested Ambient IoT device. The details are same with step2 and step3 in Figure2.

Step 3. Authentication and authorization procedures are needed between the Ambient IoT device and CN to ensure data security. However, it is worth mentioning that the matter will be deliberated and determined at SA2 and SA3

Step 4. The Ambient IoT device sends the command response to the reader if needed. For instance, if the command is enabled or disabled, in order to decrease the signaling overhead, the command response can be ignored. Another example, if the command is read, it is necessary to send the read information to the reader, which also carries the Ambient IoT device information.

Step 5. The reader transmits the command answer to the CN based on the received response message from the Ambient IoT device. It is noted that some information in the inventory response message may be visible to the reader.

B. Random access signaling procedure

During the Inventory-only and Inventory-Command procedures for DO-DTT and DT traffic, the Ambient IoT device initiates a random access operation upon receiving the initial trigger message, such as an inventory or command request. 3GPP has devised two types of random access methods for Ambient IoT: contention-based access (CBRA) procedure and contention-free random access (CFRA) technique.

CBRA procedure is widely used to the case in which a group of devices are triggered by the reader to initiate the access procedure. Figure 4 shows the CBRA procedure for Ambient IoT devices, which consists of the following steps.

Msg 0: The reader sends the triggering message to a group Ambient IoT device, which is used to facilitate to perform access procedure. The scheduling information(e.g., time or frequency occasion) for data transmission is also include in the message. Moreover, the triggering message can also named paging-like message.

Msg 1: Ambient IoT device randomly generates a random access ID and transmits it to the reader. This transmission occurs inside a single defined slot.

Msg 2: If the reader only receives a unified random access ID, it signifies that no other Ambient IoT device wants to utilize the resource at that time. The reader can send the same

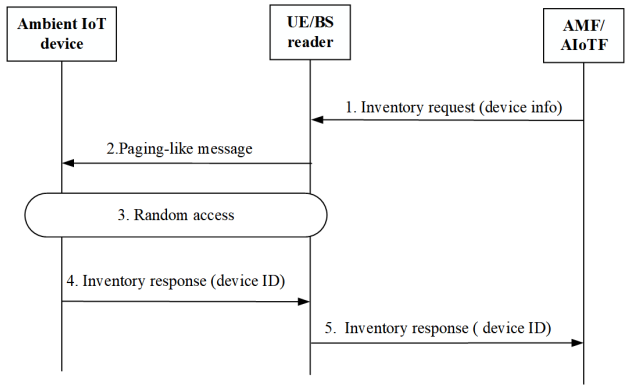


Fig. 2. Overall signaling procedure of inventory-only

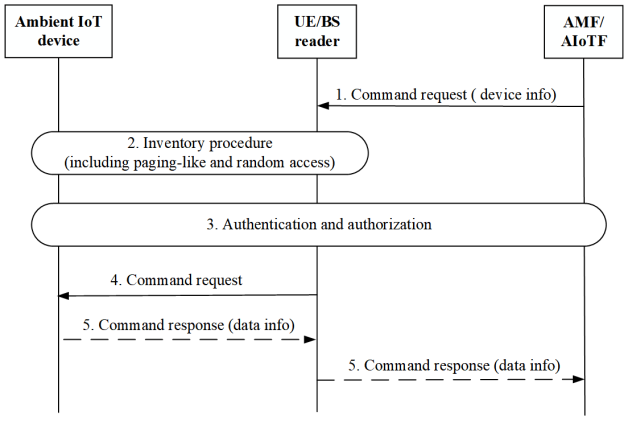


Fig. 3. Overall signaling procedure of inventory and command

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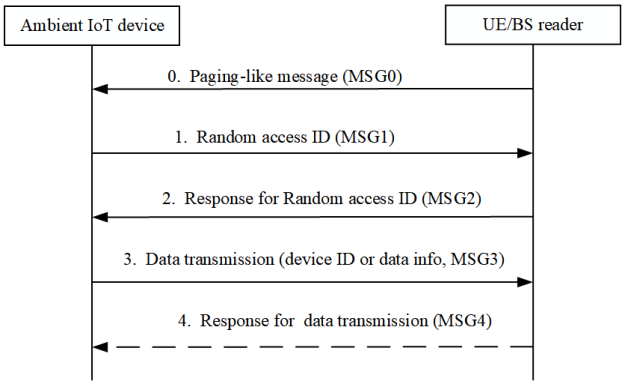


Fig. 4. CBRA procedure

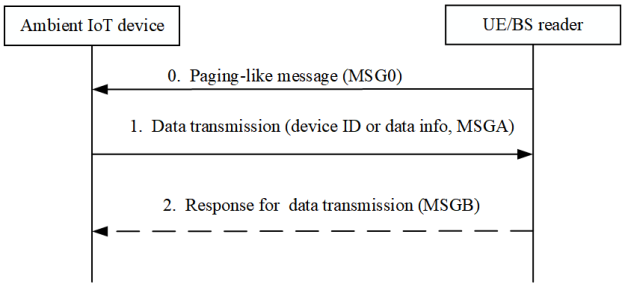


Fig. 5. CFRA procedure

random access ID or an ACK to the Ambient IoT device. Furthermore, the Msg2 may also include a resource allocation for the subsequent data transmission.

Msg 3: If the Ambient IoT device successfully receives Message 2, it will send its device ID or other relevant data information to the reader. other relevant, the Ambient IoT device will not send its information.

Msg 4: If the reader successfully receives Msg3, it has the option to transmit an ACK, which may also provide additional data for the subsequent scheduling.

CFRA procedure is widely used to the case in which one or several identified devices are triggered by the reader to initiate the access procedure. Figure 5 depicts the CFRA procedure for Ambient IoT devices, comprising the following steps.

Msg 0: The reader sends the trigger message to one or several Ambient IoT devices, including the device ID and the scheduling information(such as time or frequency occasion) for data transmission. Furthermore, the triggering message can also be referred to as a paging-like message.

Msg A: Once the A-IoT device correctly decodes the trigger message, the Ambient IoT device sends the device ID and other data information to the reader directly, depending on the assigned occasion.

Msg B: If the reader verifies the information in Msg A, it can choose to transmit an ACK or NACK to the Ambient IoT device.

V. Conclusion and Future Work

Ambient IoT technology is widely recognized as an innovative advancement in wireless communication. This paper provides a concise overview of the Ambient IoT-enabled cellular network in 3GPP Release 19, which provides the technical background of Ambient IoT. Meanwhile, it provides a detailed analysis of the essential functions of the protocol stacks used in Ambient IoT. In addition, the potential overall signaling procedures for Inventory-only and Inventory-Command cases, as well as the identification of two types of random access processes. Since the study item is still in its earliest stage, further discussions will be held in the future about specifics such as the type of device ID used over the air interface (e.g., temporary device ID, permanent ID), the MAC PDU structure of Ambient IoT, and resource allocation. Furthermore, it is important to consider whether the information of the Ambient IoT device is visible to the reader. In summary, the ongoing efforts and thorough investigations in the field of Ambient IoT offer to lay a basis for the Release 19 and forthcoming 6G network.

References

[1] Ericsson, "Ericsson Mobility Report", 2023. [Online]. Available: https://www.ericsson.com/49dd9d/assets/local/reports-papers/mobility-report/documents/2023/ericsson-mobility-report-june-2023.pdf.

[2] F. Sabrina, "Blockchain and Structural Relationship Based Access Control for IoT: A Smart City Use Case," *2019 IEEE 44th Conference on Local Computer Networks (LCN)*, Osnabrueck, Germany, 2019, pp. 137-140.

[3] Y. Wu, H. -N. Dai, H. Wang, Z. Xiong and S. Guo, "A Survey of Intelligent Network Slicing Management for Industrial IoT: Integrated Approaches for Smart Transportation, Smart Energy, and Smart Factory," *IEEE Communications Surveys & Tutorials*, vol. 24, no. 2, pp. 1175-1211, Second quarter 2022.

[4] R. Ruchi, A. Savant, A. Kalam, Y. Khurana, P. Prachi and S. Kumar, "Energy Harvesting For IoT Applications," *2022 3rd International Conference on Electronics and Sustainable Communication Systems (ICESC)*, Coimbatore, India, 2022, pp. 418-422.

[5] D. Kularatna-Abeywardana, A. Qin and H. Hu, "Ambient Environmental Energy Harvesting for Supercapacitor Driven Battery-less IoT Devices," *2023 IEEE Fifth International Conference on DC Microgrids (ICDCM)*, Auckland, New Zealand, 2023, pp. 1-6.

[6] T. Zhang, Changqiao Xu, Yibo Lian, Haijiang Tian, Jiawen Kang, Xiaohui Kuang, Dusit Niyato, "When Moving Target Defense Meets Attack Prediction in Digital Twins: A Convolutional and Hierarchical Reinforcement Learning Approach," *IEEE Journal on Selected Areas in Communications (JSAC)*, vol. 41, no. 10, pp. 3293-3305, Oct. 2023

[7] N. Mangalvedhe, R. Ratasuk and A. Ghosh, "NB-IoT deployment study for low power wide area cellular IoT," *2016 IEEE 27th Annual International Symposium on Personal, Indoor, and Mobile Radio Communications (PIMRC)*, Valencia, Spain, 2016, pp. 1-6.

[8] S. N. K. Veedu et al., "Toward Smaller and Lower-Cost 5G Devices with Longer Battery Life: An Overview of 3GPP Release 17 RedCap, *IEEE Communications Standards Magazine*, vol. 6, no. 3, pp. 84-90, September 2022.

[9] C. -H. Kang, W. -S. Lee, Y. -H. You and H. -K. Song, "Signal Detection Scheme in Ambient Backscatter System With Multiple Antennas, *IEEE Access*, vol. 5, pp. 14543-14547, 2017,

[10] 3GPP TR 22.840, "Study on Ambient power-enabled Internet of Things," V18.1.0, Mar. 2022.

[11] 3GPP TS 23.369, "Service requirements for ambient power-enabled IoT; Stage 1," V19.2.0, Jun. 2024.

[12] 3GPP TR 23.700-13, "Study on Architecture support of Ambient power-enabled Internet of Things," V19.2.0, Jun. 2024.

[13] 3GPP TR 38.848, "Study on Ambient IoT (Internet of Things) in RAN," V18.0.0, Sep. 2023.

[14] 3GPP RP-240826, "Study on solutions for Ambient IoT (Internet of Things) in NR," Mar. 2024.

[15] 3GPP TR 38.769, "Study on solutions for ambient IoT (Internet of Things)," V19.0.0, Feb. 2024.

[16] 3GPP TS 38.300, "NR; NR and NG-RAN Overall Description; Stage 2" V17.0.0, Mar. 2022.

control and user plane

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